

Junho Peter Whang

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Citizenship: Canada

Academic Positions

Seoul National University , Seoul, Korea Assistant Professor, Department of Mathematical Sciences	Aug. 2021–present
Princeton University , Princeton, NJ, USA Visiting Fellow, Department of Mathematics (sabbatical)	Sep. 2025–Jun. 2026
Massachusetts Institute of Technology , Cambridge, MA, USA C.L.E. Moore Instructor	Jul. 2018–Jun. 2021

Education

Princeton University , Princeton, NJ, USA Ph.D. in Mathematics. Advisor: Peter Sarnak. Thesis: <i>Diophantine Analysis on Moduli of Local Systems</i> .	2018
Queen's University , Kingston, ON, Canada B.Sc. in Mathematics, Honours with Distinction.	2013

Research Interests

Number theory (Diophantine analysis), arithmetic geometry, topology, and dynamics.

Publications and Preprints

1. *On indefinite integral ternary quadratic forms*.
With Alexander Gamburd, Amit Ghosh, and Peter Sarnak. Preprint, arXiv:2603.05849.
2. *Degree gaps in algebraic interpolations*.
Bull. Korean Math. Soc. **63** (2026), no. 1, 227–234.
3. *A reciprocity law on the one-holed torus*.
J. Korean Math. Soc. **62** (2025), no. 6, 1525–1538.
4. *On finite categories of algebraic varieties*.
J. Algebra **681** (2025), 206–217.
5. *Stokes matrices and exceptional isomorphisms*.
With Yu-Wei Fan. Math. Ann. **390** (2024), 4041–4086.
6. *On periodic orbits of polynomial maps*.
Preprint, arXiv:2305.13529.
7. *Surface group representations in $\mathrm{SL}_2(\mathbb{C})$ with finite mapping class orbits*.
With Indranil Biswas, Subhojoy Gupta, and Mahan Mj. Geom. Topol. **26** (2022), no. 2, 679–719.
8. *The rank two p -curvature conjecture on generic curves*.
With Anand Patel and Ananth Shankar. Adv. Math. **386** (2021).
9. *Arithmetic of curves on moduli of local systems*.
Algebra Number Theory **14** (2020), no. 10, 2575–2605.
10. *Nonlinear descent on moduli of local systems*.
Israel J. Math. **240** (2020), 935–1004.

11. *Global geometry on moduli of local systems for surfaces with boundary.*
Compos. Math. **156** (2020), no. 8, 1517–1559.
12. *The uncertainty principle and a generalization of a theorem of Tao.*
With M. Ram Murty. Linear Algebra Appl. **437** (2012), no. 1, 214–220.
13. *Another proof of the infinitude of the prime numbers.*
Amer. Math. Monthly **117** (2010), no. 2, 181.

Grants and Research Funding

Samsung Science and Technology Foundation 2022–2027

Analysis of Diophantine Problems with Nonlinear Symmetry. SSTF-BA2201-03.

Principal Investigator. Total project award: KRW 755 million.

National Research Foundation of Korea 2025–2028

Dynamics and Arithmetic Structures on Manifolds (DASOM). RS-2025-02293115.

Co-investigator. Total project award: KRW 500 million.

Selected Honors and Awards

Teaching and Learning Award, Massachusetts Institute of Technology 2021

Bershadsky Family Fellowship in Mathematics, Princeton University 2017–2018

NSERC Postgraduate Scholarship PGS D 2014–2017

Research Supervision

Ph.D. students

- Seong Youn Kim, 2022–present; co-advised with Gye-Seon Lee.
- Young Joon Ley, 2022–present.
- Eungchan Kim, 2025–present.
- Jihye Oh, 2025–present.

Master’s students

- Juhan Kim, M.S. 2023.
Thesis: *Compactification of Character Varieties via Triangulation on Surfaces.*
- Eunju Shin, M.S. 2024.
Thesis: *Nonlinear Reduction for Two Types of Generalized Markoff Equations.*
- Jihye Oh, M.S. 2025.
Thesis: *Decidability of Mutation Equivalence for Three-Vertex Quivers.*

Teaching

Seoul National University 2021–present

Graduate: Algebra, Algebraic Geometry, Commutative Algebra, Diophantine Geometry.

Advanced Topics in Number Theory.

Undergraduate: Linear Algebra, Introduction to Diophantine Analysis, Analytic Number Theory.

Massachusetts Institute of Technology

Introduction to Arithmetic Geometry (18.782). Fall 2019